

Uncovering Hidden Threats: Automated, Machine Learning-based Discovery & Extraction of Cyber Threat Intelligence from Online Sources

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Abstract—The cyber-threat landscape is constantly and rapidly expanding, overwhelming human analysts in their effort to keep track of the latest threats. This affects both organisations that produce threat intelligence to be consumed by third parties, but also the end consumers of this threat intelligence, who want, for example, to configure proactive defences to protect their infrastructure. This paper presents a novel, Machine Learning-based, solution able to discover & ingest Cyber Threat Intelligence (CTI) data from unstructured online sources, such as dark web forums, social media and online chatrooms, producing a stream of standardised, structured STIX CTI data at its output. Further, a proof-of-concept is developed and assessed, to investigate its effectiveness with real-life data sources, but also to offer insights into the large amount of potentially useful threat intelligence-relevant information that lies unused in online sources, and the positive impact that the discovery and structuring of this information in a standardised, easily shareable manner can have in terms of providing cyber defenders with an up-to-date and comprehensive view of the threat landscape.

I. INTRODUCTION

The cyber-threat landscape is vast, and it has become inefficient for humans to survey [1] [2]. Cyber risks emerge from every new hastily produced software update, every bit of hardware that goes out to the production line while cutting corners in its security, and every new technologically-enhanced application that has not been thoroughly assessed before being offered to end users. This affects everyone, from individual users to big corporations and government institutions. Most of the cyber-security efforts focus on reactive actions, after an event has occurred [2], even though the benefits of a more proactive approach have been extensively argued [1] [2].

Cyber Threat Intelligence (CTI) has emerged in recent years as a key enabler towards proactive defense strategies. While a new category of products and services has spawned around CTI (e.g., CTI feeds & platforms), even more novel approaches focus on leveraging Artificial Intelligence (AI), and Machine Learning (ML) in specific, to autonomously scrape heterogeneous online sources, towards the more timely discovery of CTI. Through machine integration, big data analysis can be integrated into the CTI pipeline and have a positive impact on the overall security of individuals and organizations. This automation can also expand the reach of standard CTI-gathering mechanisms exploiting information from hacker

blogs and forums, which include a priceless glimpse into the inner workings of relevant underground communities [3].

A. Research Gap

There is a considerable effort going on worldwide to automate CTI gathering and analysis; however, existing works that are fully automated are still in their infancy. Most of the available solutions, either combine a human-in-the-loop approach in the information chain [1] or the information itself relies on human-made machine-readable CTI data [2]. The latter are typically generated by large teams of expert CTI analysts, and are made available via a subscription fee. Existing efforts aiming to tackle this problem, both from academia and the industry, are further discussed in Section II.

Overall, by minimizing the CTI gathering and processing time, threat response time and effectiveness can be significantly increased. This development allows predictions to be made about future events and attacks, allowing organizations to enhance their preparedness against such incidents. Per Gartner's 2023 report on the Threat Intelligence (TI) market, "*successful TI services have risk assessment capabilities associated with threat actor groups, tactics, techniques and procedures (TTPs), indicators of compromise (IOCs), exploits and others*" [4].

Recent efforts shift their focus on Machine and Deep Learning methods to deal with the large amount of user-generated data on the internet [5], as the huge amount of available unstructured data necessitates the employment of such tools and methods, with CTI being no exception.

B. Overarching Objective

Motivated by the above, this paper introduces a novel solution, referred to as CTI-DATH, that can discover and ingest CTI data from unstructured online sources (e.g., dark web forums, social media, online chatrooms) and automatically produce a stream of standardised, structured CTI data, without human interaction. The workflow of CTI-DATH can be seen in Fig. 1.

In more detail, the proposed solution uses intelligent crawling and scraping to gather unstructured CTI data from the web (e.g., malicious forums, security blogs, as well as social media, like Twitter/X), extracting useful information from relevant

accounts [6]. Machine Learning, Natural Language Processing (NLP) and Semantic Analysis techniques are employed to process the collected data into a structured format in order for the output data to be understandable and actionable, both for humans and machines. More specifically, the standardized data format that was chosen for CTI-DATH is STIX 2.1 [20] by the OASIS Foundation [19], which also minimizes the integration time of the proposed solution with other CTI management tools.

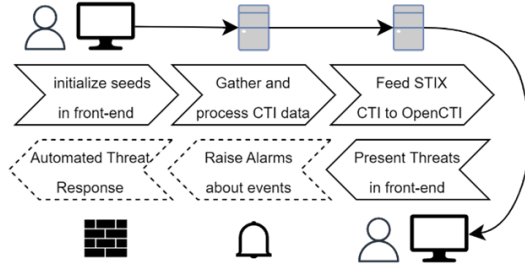


Fig. 1. The CTI-DATH workflow.

In summary, the contributions of this paper include:

- The area of unstructured CTI-relevant data present on the web is explored.
- The CTI-DATH tool both as an architecture and as an implementation, enabling automatic discovery and extraction of CTI from heterogeneous online sources, combining State-of-the-Art methods, such as ML, NLP & Convolutional Neural Networks (CNNs).
- A PoC of the proposed solution is presented, it's results evaluated, and discussed.

To present the above, the paper is organized as follows: Section II presents the State-of-the-art in the field; Section III details the proposed approach; Section IV presents the PoC, while Section V discusses the evaluation results; Section VI outlines the conclusions and future research directions.

II. RELATED WORKS

According to Gartner's 2023 Report on the CTI Market, although more organizations opt for TI products to safeguard their cyber-space, most do not have a formalized and automated process, resulting in a compromised ability to proactively protect their assets [4]. This has led researchers to create solutions for CTI discovery and categorization, many of which are limited to manually studying downloaded data from websites [1], or using partially automated tools [7]. There is also some research on CTI classification from large existing datasets, produced from whole-forum dumps [2].

A summary of related works can be found in Table I, the more relevant of which are presented below.

Sun et al. [9] proposed an architecture that deals with the collection, processing, classification of articles, and their conversion to a CTI standard format, like STIX. The researchers collect information from Open-Source Threat Intelligence Platforms (OSTIPs), classify the CTI and, after deciding

TABLE I
PRIOR RESEARCH ON AUTOMATED CTI DISCOVERY AND ANALYTICS.

Research	Forums	Social Media	Other	STIX
Sun et al [9]	No	No	Yes	Yes
AttackKG [12]	No	No	Yes	No
SECDFAN [8]	Yes	N/A	N/A	Yes
STIXnet [7]	No	No	Yes	Yes
TTPDrill [10]	N/A	N/A	Yes	Yes
SYNAPSE [6]	No	Yes	No	No
EXTRACTOR [13]	No	No	Yes	No

if they contain useful information, they map them to STIX through precise mapping rules. They trained their models on existing TTP knowledge bases, like MITRE's ATT&CK. The main limitation of this implementation is that they generate only 4 of 18 types of STIX Domain Objects (SDOs). The overall measured and evaluated accuracy of their method was above 90%.

Sakellariou et al. [8] proposed the *SECDFAN* architecture, which can extract raw data from CTI forums automatically or manually and convert them to STIX data. The researchers split their architecture into layers: one for source selection, one for surveillance (crawlers), one for processing (feature extraction, correlation etc.), and finally one for STIX object creation. The authors thoroughly presented the requirements and architecture of *SECDFAN*; however, they did not provide any insights about the implementation (e.g., a PoC) or evaluation of this approach.

Marchiori et al. [7] proposed and implemented an analytical pipeline called *STIXnet* to convert CTI Reports to STIX objects, employing manual data gathering. The researchers use NLP and ML models trained on ATT&CK and Deep Learning to extract entities of interest, like Indicators of Compromise (IoCs), known and new CTI entities, and relationships between them. They developed a platform that can create STIX ontologies from CTI reports. However, a drawback is that *STIXnet's* performance has not been tested on extracting non-structured text from forums and social media.

Husari et al. [10] proposed a similar architecture in *TTPDrill* for extracting Tactics, Techniques and Procedures (TTPs) from online blogs and articles. This work demonstrates key concepts like data sanitization, and TTP similarity identification to improve detection accuracy. It was evaluated mainly on Symantec CTI reports, which share a great similarity in structure and language use. Therefore, it can be considered very narrow in applicability scope.

A solution named *SYNAPSE* that aimed at social media like X, was proposed by Alvez et al. [6]. It collects tweets from X through the official API, without dealing with scraping. *SYNAPSE* performs feature extraction and classification to get MISP objects. Nevertheless, a main limitation of this work is that it focused only on one source (i.e., tweets from X) gathering data from the official API, without going deeper and scraping the whole website.

Satvat et al. [13] proposed a similar architecture, named *EXTRACTOR*. The authors employed advanced NLP tech-

niques to advance the information that can be extracted from CTI reports to capture more details. The outcome of the *EXTRACTOR* is a provenance graph, where the nodes are the entities that participate in and are influenced by the attack and the edges are the system calls that connect them. Even though, the *EXTRACTOR* accomplished very good results in extracting information from unstructured CTI reports, it did not classify them in a well-known standardized format (e.g., STIX 2.1).

To address the limitations of existing works, CTI-DATH aims to minimize human involvement in the CTI pipeline (collection, processing, and dissemination of CTI) by integrating and combining various open-source tools and state-of-the-art deep learning and NLP methods. Therefore, human involvement is limited only to supervisory and initialisation tasks. Moreover, a PoC of CTI-DATH is presented to demonstrate the efficacy of the proposed solution in a realistic implementation.

Still, some of the prior works mentioned above influenced or were partially integrated within CTI-DATH, as presented in Table II and are also detailed later in the manuscript.

TABLE II
THE OPEN SOURCE AND ACADEMIC TOOLS USED IN CTI-DATH.

Project	Main Features	Comment
STIXNet [7]	STIX Objects creation	Recognizes novel entities
ACHE [23]	Focused web crawler	From VIDA Lab, NYU
Pdf2htmlEX	PDF to HTML	Open-source project
IOC Finder [24]	IOC Extraction	Extracts IOCs from text
rcATT [26]	ATT&CK TTP output	Fairly old, outdated
deepdarkcti [25]	Dark Web sources list	Extensive list of sources
TRAM [27]	LLM TTP Identification	Supports ATT&CK
OpenCTI [22]	CTI Management	Ingests/Disseminates CTI

III. THE PROPOSED ML-BASED CTI DISCOVERY & CLASSIFICATION TOOL

In this section, we elaborate on the internal characteristics and components of the proposed solution.

A. Proposed Architecture Overview

The proposed architecture, as illustrated in Fig. 2, consists of several main components designed to navigate, scrape, process, and disseminate CTI data in a standardised format.

The tool, which we named CTI-DATH, includes a *crawler* that explores websites, seeking new links based on predefined regular expressions while avoiding previously accessed locations. The crawler provides a list of URLs to a *scraper*, which extracts the website data and saves them locally.

Following the data extraction, the *preprocessor* dissects the HTML into separate entities (e.g. single forum posts), extracts the human-generated text, and cleans it. A *decider* module then evaluates whether the extracted data is security-related and valuable as CTI. The *db manager* receives input from the *decider* to store the raw extracted data. A core component of CTI-DATH is the *Entity eXtraction Module* (EXM) that identifies and extracts various STIX Domain Objects (SDOs) and STIX Relationship Objects (SROs) using Natural Language Processing (NLP), Machine Learning (ML) and knowledge base Named Entity Recognition (KB NER). This module

includes a Natural Language Processor for entity extraction, a CTI Quality Estimator for assessing the trustworthiness of the information, and a tf-idf classifier to categorize the SDOs.

The extracted information is then fed to the STIX mapper to generate the STIX objects and ontologies based on the entities identified by the EXM component. To facilitate the sharing of CTI data, the architecture incorporates an *OpenTAXII* instance, a server designed to handle the digestion and dissemination of STIX data.

Lastly, the *TAXII Client* interfaces with the *OpenTAXII* server to store or disseminate data processed by the STIX mapper.

B. Core Components

The proposed solution is based on a set of components that are independent from one another to enhance its modularity. The individual components of CTI-DATH are detailed below.

Crawler: A Crawler program was utilized to navigate websites of interest using initial seeds (e.g., known hacker forums, see [25]) and a snowballing technique [1]. It accesses dark web security forums, PDF reports, and social media like Twitter, which is useful for obtaining current security threats [6]. Proxy servers are needed to obscure its identity from anti-bot rules. The ACHE open-source project [23] was used for this purpose.

Scraper: The scraper component receives input from the crawler, and specifically the URLs to gather content in a secure environment (e.g. in our case, a virtual machine) [1]. It extracts the main content (HTML, hyperlinks etc.) and metadata useful for CTI analysis.

Decider: The Decider module uses a tf-idf model that is based on a local collection of a variety of cybersecurity related keywords (e.g. malware names, threat actors etc.) to distinguish random internet texts from the ones that are of interest.

Preprocessor: The preprocessor extracts the human text from the HTML, omitting unnecessary parts. It does that by removing HTML tags and splitting data into plaintext, code snippets and attachments, similarly to what proposed by Samtani et al. [14].

Natural Language Processor: The NLP component processes the text by performing the basic NLP cleaning techniques(tokenization, stop word removal etc.). This is a crucial step to remove unwanted noise in the data later on the processing chain. The OpenNLP Java library was used for this task.

Entity Extraction Module: This module extracts entities of interest using techniques ranging from Regular Expressions for common IoCs and Named Entity Recognition (NER) for novel objects. It also employs Knowledge Based Named Entity Recognition (KB NER) to spot occurrences of known entities.

CNN CTI Classifier: The main component of CTI-DATH, the CTI Classifier is based on CNNs to classify the unstructured data into various CTI classes. This method was chosen due to its ability to classify complex data with high accuracy. The CTI classes which were used for the classification are based on the STIX 2.1 ontology [20].

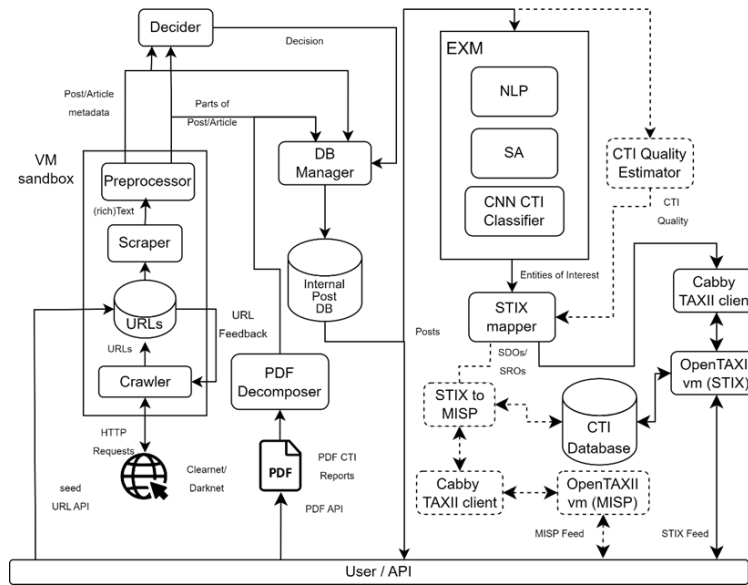


Fig. 2. The CTI-DATH Architecture

STIX Mapper: The STIX mapper creates the data fields of each STIX Domain Object (SDO) using inputs from previous stages, directing output to the OpenTAXII server for storage and dissemination.

OpenTAXII Server: The OpenTAXII server has been employed to exchange CTI via HTTPS. OpenTAXII is an open-source solution that has been implemented to support both exchange of STIX objects, although it is context agnostic [21]. It is important to mention that such TAXII server can be integrated with any CTI Management System, such as OpenCTI, to visualize, analyze and act on the gathered data.

IV. PROOF OF CONCEPT IMPLEMENTATION

A. Testbed Details

The Proof Of Concept assessment was performed on a test server, deployed on a laptop with 32GB of memory, an Intel i9-13900H and a Micron 2400 NVMe (See Fig. 3 for a screenshot of the command line interface of the tool).

To develop the PoC, a Java service and a set of dockerized services able to run simultaneously have been deployed. These services include a full OpenCTI installation, an OpenTAXII fork server¹ which was adapted to be compatible with STIX 2.1, and a VIDA/NYU Ache Crawler server instance [23].

The project requires a set of services to run (Elasticsearch, Kibana, ACHE Crawler, OpenTAXII and optionally, OpenCTI), so the resource demands are equally significant. Still, the deployment has low complexity, since these services are part of a custom dockerfile, and they can be hosted anywhere if the user ensures that the ports the use of the above services are forwarded and open.

¹<https://github.com/eclecticiq/OpenTAXII>

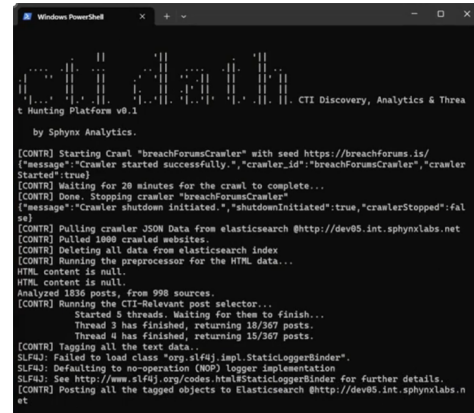


Fig. 3. The tool's command line interface

B. Tool Workflow

CTI-DATH is configurable by the user to cycle through a set of pre-defined chains of actions, in specific time intervals. This results in a near-constant feed of new CTI objects. It is worth noting that only a very small percentage of online content we target can result in CTI data, typically much less than 1%. This creates the need for long cycles that may last hours just to produce a single-digit number of actionable data. The cycle of operation for the PoC of CTI-DATH is illustrated in Fig. 4.

Crawling: The crawling process is the first step to gathering generic information from the web and dark web. It runs for a specified amount of time or a number of web locations. This process stores every web page it crawls in the Elasticsearch database that the OpenCTI tool already has.

Preprocessing: The preprocessor recovers all the saved data after the crawling phase is finished, and extracts the human

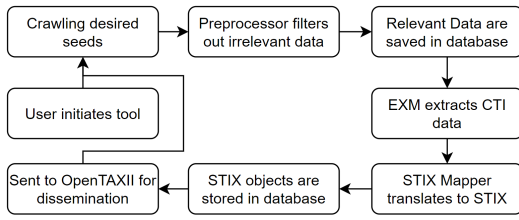


Fig. 4. The PoC cycle of operation

text of interest from each webpage.

Scraped Data Storage: The preprocessed data is forwarded to a different database, to be stored for future reference.

CTI Extraction: The EXM module extracts all the possible STIX objects, by tagging the text provided by the previous phase.

STIX Mapping: The STIX mapper creates STIX 2.1 - compliant data in JSON Format, based on the received tagged text from the CTI extraction process.

STIX Data Storage: The STIX 2.1 objects are stored in the same Elasticsearch database of OpenCTI, as the webpage content is both easily integratable with other tools and for the storage is non-volatile.

STIX Data Dissemination: When all the above processes are concluded, the outcome is a database index that contains newly created STIX data. These are sent to the OpenTAXII server to be disseminated as an STIX feed, which in our PoC setup is ingested by an OpenCTI instance.

Finally, when the cycle is over, it starts over, until the process is killed by the user, or the user-specified crawling limit is reached.

V. EVALUATION & DISCUSSION

Through extensive testing of the PoC, it has been validated that CTI-DATH can identify CTI-related pieces of text in pools of unstructured textual data, leveraging its Decider module. This is based on crawls of 312 pages from the (now-FBI-confiscated due to possible cyber-criminal content) BreachForums² dark web forum performed on February of 2024, and manual tagging of 698 posts. Also, the tool proved it can recognize several entities mapped to STIX SDOs in human-generated text.

A. Text Classification Accuracy

Given that the relevant text identification is based on the frequency of words used in CTI-related content, the PoC presented moderate success in distinguishing the cyber-security-related posts/threads. The main drawback lies in the countless non-human-generated posts found online, which are either ads or announcements and contain many cybersecurity-related words, but do not represent valid human sentences. The Decider module correctly recognized CTI-related text 4.7% of the time, depending on the parameters it was given for each source (forums, posts etc.), leaving a 2% of false positive

²<https://breachforums.is>

TABLE III
RESULTS OF KNOWLEDGE BASE AND CONVOLUTIONAL NEURAL NETWORK NER FOR CLASSIFYING TEXT INTO STIX SDOs.

Entity(STIX SDO)	Method	Accuracy	False Positives
Attack Patterns(known)	KB NER	99%	1%
Threat Actors(known)	KB NER	99%	1%
Infrastructure(known)	KB NER	99%	1%
Tool(known)	KB NER	99%	1%
Malware(known)	KB NER	99%	1%
Location (novel)	CNN NER	71%	9%
Identity (Organ/ion) (novel)	CNN NER	65%	7%
Identity (Person) (novel)	CNN NER	43%	5%
Note (Date/Time)	CNN NER	90%	1%

results, and a 0.6% of failing to identify text of interest. The amount of false positive results indicates that there is a need for further research and experimentation in this area using more advanced ML and DL methods, such as BERT, and LLMs [16] [18].

B. Named Entity Recognition Accuracy

Some of the entities, such as some STIX Cyber Observables are recognized using groups of regular expressions. Others, like a sub-group STIX SDOs, are matched with local Knowledge Bases (KB) from reputable sources such as MITRE ATT&CK, STIX2.1 Vocabularies in a process we call Knowledge Base Named Entity Recognition (KB NER). This matches the combination of words from the input text to a database of known entities, based on word similarity. This method has an excellent detection rate, and an extremely low false positive rate which is only due to common words being used sometimes to name such entities (e.g. the "Zeus" malware). Finally, some of the novel objects are recognized individually using Named Entity Recognition (NER) and CNN implementations of OpenNLP libraries. Currently, there is no support for sentence recognition as a single entity, but it has been identified as a direction for future research. Table III represents the experimentally measured accuracy of each entity recognition, with the subset of 100 posts of the dataset described in the previous section.

Overall, the assessment of the PoC highlights the need for more advanced algorithms that can recognize entities based on deep, multi-layer textual context other than the word and the sentence itself. The above represents a lightweight approach that can be applied to real-world scenarios, but it is also one that can accept further research and upgrades.

VI. CONCLUSIONS & NEXT STEPS

The work presented herein managed to reach most the goals it had initially set out to achieve. The comprehensive architecture that was defined, was followed by the development and assessment of an actual PoC tool, revealing some practical limitations in current tools and methods, such as pipeline complexity, significant resource needs, and the long-term maintenance of a modular architecture that integrates the latest, albeit heterogeneous, SoTA tools and methods. Such

insight would not have been revealed by theoretical analysis alone, or by studying the individual tools in isolation.

In terms of future work, the CTI-DATH tool itself (still at a proof-of-concept stage) can be greatly improved in terms of functionality & usability. Other planned improvements include:

- Integration with more sources, like IRC channels, Telegram channels etc.
- Enhancement of the extraction of STIX 2.1 SDOs by improving the models used for Named Entity Recognition and text classification, either with the addition of training data or by using other, more efficient models.
- Addition of datasets with CTI Reports and their corresponding STIX 2.1 representation, in the model training pipeline.
- Software-level changes, like introducing a user-friendly UI and APIs.
- Evaluating the replacement of the Entity Extraction Module internals with Large Language Models (LLMs) [16] [17] [18].

Overall, the results of this work re-enforce the position that autonomous CTI collection from human-readable online sources is a research (and potential business) sector that offers a lot of new opportunities for strengthening the cyber-security of organizations, and can be leveraged to gain access to information sources previously inaccessible, overall improving the capacity for proactive cyber-defence.

Further, the above will be validated in the context of realistic use cases, involving operators of essential services (such as energy, transport, and healthcare), in the context of EU-funded projects PHOENIX [28] and SYNAPSE [29].

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